When Control Corrupts Information

Feasible Auditing, Report Use, and the Nonmonotone Demand for Protection

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Abstract

How should organizations combine independent audits with the use of employee reports for evaluation? We solve a finite-menu governance problem in which workers separately prevent defects, discover them and disclose findings. Audits verify occurrences but cannot prove prior awareness, excluding concealment penalties from the implementable benchmark. At fixed incidence, audits have no direct discovery or disclosure effect. Jointly optimizing audit intensity and report use yields a globally certified nonmonotone protection policy: higher information value can reduce, then increase, report use before full protection becomes optimal. Total optimized information remains monotone in its value. Complete proofs, boundary cases and synthetic interval certificates establish the result within the constrained menu; no empirical discovery or dishonesty effect is identified.

1. Introduction

Operational records are both inputs to learning and evidence used to evaluate the people who produce them. An organization choosing a reporting policy must consider prevention, discovery, disclosure, the independent information produced by audits, and the resource cost of protecting a learning channel. A higher report count is not automatically better information, and a lower count is not automatically lower operational risk.

This paper studies a restricted but implementable policy-design problem. The organization chooses an audit intensity and a report-linked evaluation charge from a finite menu. A contractual liability cap rules out some combinations. Audits can establish a defect and sometimes its diagnostic context; they cannot establish whether an employee previously knew about it. No penalty is imposed merely by labeling an omitted report deliberate concealment. Individual report use can still discourage disclosure, because a submitted report is observable. Protection means reducing this report-linked charge, not abolishing independent accountability.

The principal result is a nonmonotone optimal demand for protection. When learning becomes more valuable, choosing a stronger audit technology can make increased individual report use optimal even though it reduces worker reporting sensitivity. Independent diagnostic information replaces some worker-produced information; saved protection costs and prevention gains can justify that substitution. This is not the claim that controls generally improve information or that concealment penalties are desirable. It occurs in the feasible benchmark with no awareness-contingent penalty.

The article supplies an exact finite-menu solution and parameter regions, rather than a comparison of two selected regimes. Finite menus represent administratively approved audit and access-control packages. Optimality is relative to that menu, a fixed occurrence penalty, and a fixed reporting reward. It is not optimality over all contracts. The upper-envelope method itself is elementary. The substantive result is a counterexample to monotone protection, obtained with endogenous prevention, discovery, disclosure, combined audit/report information, and enforceability constraints. Its importance outside this model remains an empirical question.

2. The predecessor and the contribution boundary

The actual author-hosted published article by Graff Zivin and Pfaff (2004), not a working version, is the closest predecessor. Its results are equation-based rather than numbered propositions:

Published location	Existing implication
Section 2.2, equations (4)-(5), pp. 941-942	Reporting increases exposure and distorts adoption despite incentives for correction.
Section 3.1, equations (6)-(8), pp. 942-943	Differentiated penalties can equate expected exposure and restore adoption incentives.
Section 3.2, equations (9)-(12), pp. 943-945	Hidden physician detection/reporting effort and agency costs already enter.
Section 4.1, pp. 946-947; Appendix A, pp. 947-948	Observable proxies and outcome-contingent incentives address unobserved effort.

Thus sanctions deterring reports, endogenous detection, and the general incentive-design problem are not new. Here the added object is joint menu choice under a liability cap, with audit-produced diagnostic signals and no inference of awareness. The nonmonotonicity result is not implied by the predecessor's stated policy equalization conditions.

The broader boundary is also important. Pfaff and Sanchirico (2000) already analyze self-investigation and correction incentives, including observability problems. Kaplow and Shavell (1994) characterize optimal self-reporting enforcement, including enforcement-resource benefits. A paper cannot claim originality merely by adding audit costs or calling report-use restrictions governance. Their primary publisher or author records were verified; those records support the broad comparison, not a claim to have exhausted every theorem in their restricted full texts.

Dai (2026), a particularly close recent comparison, jointly optimizes regulatory audit intensity and penalties under limited liability. Its Lemmas 1-2 concern self-audit timing; Proposition 1 characterizes regulatory policies. The present article therefore does not claim joint audit/penalty design or audit crowd-out as new. Its distinct target is how the value of diagnostic learning changes optimal report-use protection when worker discovery and disclosure are separate.

Wang, de Véricourt, and Sun (2024) solve a richer dynamic audit/remediation problem with evasion; the consulted author manuscript's Theorem 1 establishes optimal prompt disclosure in its contract space. That result reinforces why our restricted-menu optimality cannot be advertised as a general optimal-contract theorem. Strausz (2006), compared using the October 2004 manuscript and verified final metadata, connects reporting, audit precision, and rents. Here audit source independence and costly suppression are different assumptions, not discoveries that reporting and auditing interact.

Bénabou and Tirole (2016) place multitasking within competitive screening. The verified final publication is distinct from the March 2013 author manuscript consulted for its framework. Edmondson (1996) and Bernstein (2012) supply direct organizational precedents on detection, learning, visibility, and concealment. Maas and van Rinsum (2013) caution that identification can sometimes improve reporting honesty. These precedents motivate allowing both information losses and benefits from accountability. The proposed contribution is narrower than discovering endogenous organizational information.

3. Worker behavior and enforceable policies

3.1 Production of defects and reports

One worker handles one item. Preventive effort $e \geq 0$ costs $c_e e^2/2$ and makes a defect occur with probability

$$p(e) = p_0 e^{-e}, \qquad 0 < p_0 < 1.$$

Investigative attention $d \in [0,1]$ costs $kd^2/2$, whether or not a defect exists, and discovers a true defect with probability d. Following discovery, the worker chooses concealment intensity $c \in [0,1]$, at real expected cost $\eta c^2/2$ per discovery. Disclosure then occurs with probability $1-c$. This reduced-form convex concealment technology represents the cost of suppressing discovered information; it is not a statement about observed employee intent. Reporting sensitivity and raw report probability are

$$q = d(1-c), \qquad r = pq. \tag{1}$$

There are no false positives in the behavioral baseline. Their empirical measurement is addressed separately. A discovery yields private immediate remediation benefit $u \geq 0$ even without disclosure. A disclosed finding earns a reward $v \geq 0$ and incurs report-use evaluation charge $s \geq 0$. A randomly selected independent audit, with probability a, imposes occurrence penalty $b \geq 0$ if a defect exists, whether or not it was reported. Let $h > 0$ be private defect harm and $H = h + ab$.

Audits occur after the worker reporting window and immediate private remediation. They do not substitute for the immediate benefit u. This timing/separability assumption is important: awareness-blind audits can directly crowd out investigation in other models, including audit-substitution settings such as Dai (2026). Theorem 2 below is a benchmark within (2), not an impossibility theorem across all feasible technologies.

An awareness-contingent penalty would be F per discovered but concealed defect caught by an audit. The general utility expression is

$$U = -\frac{c_e e^2}{2} - Hp + pd\left[u + (v-s)(1-c) - \frac{\eta c^2}{2} - aFc\right] - \frac{kd^2}{2}. \tag{2}$$

The governance analysis imposes $F = 0$. Occurrence alone does not prove awareness. A report-use charge is enforceable because submission itself is observable; an occurrence penalty relies on the independent artifact. Charges are assumed contractual monetary transfers, not uncontrolled reputational or career damage. Fixed compensation satisfies participation. This assumption permits a coherent inclusive-surplus objective and restricts its organizational interpretation.

3.2 Feasible menu and liability

The organization commits before worker choices to a deterministic pair (a, s) in a finite candidate set $\mathcal{A} \times \mathcal{S}$. Auditing capacity bounds $a \leq \bar{a} \leq 1$. A gross per-item liability cap B requires

$$s + b\mathbf{1}\{a > 0\} \leq B. \tag{3}$$

If an audit can occur, an audited and reported defect can incur both charges; if $a = 0$, only the report charge can occur. Equation (3) caps gross negative transfers before the separate reporting reward. It is a contractual cap, not an estimate of a jurisdiction's liability law. No policy may charge an unobserved awareness fine. Audit probability is selected before seeing reports and is not targeted by their contents.

Real audit cost is $K(a) = \kappa a^2/2$, $\kappa > 0$. Reduced report use requires access restrictions and enforcement of a use boundary, with real channel cost $C(s) = \gamma\phi(s)$, $\gamma > 0$, where

$$\phi(s) = [\max\{1 - s/\bar{s}, 0\}]^2 . \tag{4}$$

Full protection is $s = 0$, not zero audit intensity. The specified cost schedule is an assumption about implementing protection. It is not a finding that every organization faces quadratic governance costs. The menu, cap, capacity, and cost schedule together define the constrained design problem. Random policy mixing is unnecessary for a linear objective over independently treated items: an average of deterministic policy surpluses cannot exceed their maximum. Hidden lotteries that workers cannot observe before choosing actions would be a different contract space.

3.3 A global worker solution, including zero information

Write $[x]_0^1 = \max\{0, \min\{1, x\}\}$ and $g_+ = \max\{g, 0\}$. For a continuation following discovery,

$$c^* = \left[\frac{s - v - aF}{\eta}\right]_0^1, \qquad g = u + (v - s)(1 - c^*) - \frac{\eta(c^*)^2}{2} - aFc^*. \tag{5}$$

For all policies under consideration assume a common bound $G \geq g_+$ satisfying

$$\frac{p_0 G}{k} < 1, \qquad h > \frac{2p_0 G^2}{k}. \tag{6}$$

These primitive bounds deliberately exclude saturated attention along the effort domain; they do not exclude zero attention or either concealment boundary. For nonnegative s, F, one may take $G = u + v$. A tighter policy-specific bound can also be used.

Theorem 1 (global response). Under (6), each policy has a unique preventive effort and investigative attention:

$$d^* = pg_+/k, \qquad c_e e = pH - p^2 g_+^2/k. \tag{7}$$

The effort equation has exactly one positive solution on $[0, \infty)$. When $g \leq 0$, attention and reporting are zero. The continuation value c^* is still well defined, but no realized concealment rate is observed if nothing is discovered. When $s \leq v + aF$, concealment is zero; when $s \geq v + aF + \eta$, disclosure is zero. Equations (5)-(7) remain valid at all these boundaries and are continuous in policy. All proofs appear in Appendix A.

4. What implementable audits do and do not do

4.1 The feasible benchmark: no direct attention reversal

With $F = 0$, c and g depend on s but not on a. Let

$$\Delta = c_e + Hp - 2p^2 g_+^2/k > 0.$$

Theorem 2 (awareness-blind audit benchmark). At fixed defect incidence p, changing a leaves discovery and disclosure unchanged, including zero attention and concealment boundaries. With endogenous prevention,

$$e_a = pb/\Delta, \qquad p_a = -pe_a, \qquad q_a = -e_a q. \tag{8}$$

Thus audit-induced reductions in reporting sensitivity in this benchmark are entirely mediated by prevention. There is no fixed-incidence audit corruption effect. If $b = 0$ or $q = 0$, the corresponding derivative is zero. One cannot reinterpret (8) as evidence that employees hide known findings when auditing increases.

At fixed a, greater report-linked charge s can increase prevention and reduce discovery and disclosure. In an interior concealment region with $g > 0$,

$$e_s = 2pd(1-c)/\Delta, \qquad q_s = -e_s q - p(1-c)^2/k - d/\eta. \tag{9}$$

The second term is the direct discovery loss; the third is disclosure conditional on discovery; the first is the indirect prevention margin. At concealment corners the derivative of c is zero locally; at $g \leq 0$, reporting is zero. These effects do not establish that full protection maximizes welfare.

4.2 Information from both sources

Each item has a latent binary root-cause label θ with prior one half, independent of whether a defect occurs. A worker report reveals the label. An audit always verifies occurrence and, independently of worker reporting conditional on a defect, recovers the label with probability $\rho \in [0, 1]$. Audit diagnostic capability is not proof of the worker's prior awareness. An audit can reconstruct a cause without reconstructing an employee's knowledge history.

Counting each defect at most once, total occurrence-detection probability and diagnostic-label arrival probability per item are respectively

$$T = p[q + a(1-q)], \qquad I = p[q + a\rho(1-q)]. \tag{10}$$

Conditional diagnostic sensitivity is $J = I/p$. Reports and audits can overlap; adding $pq + pa\rho$ without subtracting overlap would overstate information. When $\rho = 1$, $I = T$; when $\rho = 0$, auditing still verifies occurrences, but diagnostic learning in this specification comes from worker reports only.

A subsequent remedy decision pays V if its chosen binary action matches θ and zero otherwise. With no label, optimal expected payoff is $V/2$; with a label, it is V. Since the absence of a label is independent of θ, incremental decision value is LI, where $L = V/2$. Any separate value assigned to occurrence verification can be added as MT, $M \geq 0$. We set $M = 0$ for the explicit example, rather than counting occurrence information as worthless in every application. The general finite-menu solution below allows $M \geq 0$.

Under Theorem 2, combined conditional diagnostic sensitivity and unconditional learning supply obey

$$J_a = \rho(1-q) - (1-a\rho)e_a q,$$

$$I_a = p\{\rho(1-q) - e_a[a\rho + 2(1-a\rho)q]\}. \tag{11}$$

At fixed incidence, $J_a = \rho(1-q) \geq 0$. With prevention, either sign of (11) is possible. A decrease in learning opportunities because fewer defects occur is not direct corruption of discovery or disclosure. The design problem must value real prevention separately from the supply of diagnostic signals.

4.3 Exact boundary of the earlier perverse-audit mechanism

For comparison only, suppose an external technology could prove awareness, so $F > 0$ were feasible. This technology is not assumed available or optimized here. At fixed p, allow the general attention cap $d = [pg/k]_0^1$, even outside (6).

Theorem 3 (primitive reversal region). Put $x = s - v$ and $t = (u - x)/\eta$. A strictly negative fixed-incidence derivative $\partial q/\partial a$ is possible only with $0 < c < 1$ and $0 < pg/k < 1$. In this region,

$$g = u - x + \eta c^2/2, \qquad \left.\frac{\partial q}{\partial a}\right|_p = \frac{pF}{k}\left[t + \frac{3}{2}c^2 - c\right]. \tag{12}$$

It is negative exactly where $g > 0$ and

$$c_- < c < c_+, \quad c_\pm = \frac{1 \pm \sqrt{1 - 6t}}{3}, \tag{13}$$

intersected with $0 < c < 1$ and reachable $c = (x - aF)/\eta$. Real distinct roots require $t < 1/6$. In particular, $u - x \geq \eta/6$ rules out reversal globally. In a saturated-attention region the derivative is $F/\eta > 0$; in the interiors of zero attention, zero concealment, or full concealment it is zero. At kinks the function is continuous and one-sided derivatives follow the adjacent formulas. Setting $F = 0$ removes the entire fixed-incidence reversal, not just its numerical example.

This result delimits a specific penalty technology. It does not recommend implementing awareness fines or claiming that an omitted report proves awareness. For example, at $p = .1$, $u = .6$, $s = .7$, $v = .1$, $a = .2$, $F = \eta = 1$, and $k = .3$, one has $c = .4$, $g = .08$, and the direct derivative is $-.05\overline{3}$. This is a synthetic comparative static of the hypothetical technology, distinct from the feasible governance results that follow.

5. Joint governance design

5.1 Objective and exact solution

Let $D \geq h$ be inclusive real harm from a current defect, excluding the private remediation gain u and future remedy payoff. Real behavioral resource cost is

$$E_j = c_e e_j^2/2 + k d_j^2/2 + p_j d_j \eta c_j^2/2.$$

For feasible policy j, define

$$B_j = -Dp_j + up_j d_j - E_j - \kappa a_j^2/2 - \gamma\phi(s_j) + MT_j.$$

The constrained design problem is

$$\max_{j \in \mathcal{M}(B)} W_j(L), \qquad W_j(L) = B_j + LI_j, \tag{14}$$

where $\mathcal{M}(B)$ consists of all candidates satisfying (3) and capacity. Charges and rewards cancel as internal transfers under risk neutrality, fixed-wage participation, and the stated accounting boundary. Real effort, concealment costs, audit costs, and protection costs remain. If charges are psychological or reputational losses, additional welfare terms are needed. No welfare conclusion is claimed for that different accounting.

Theorem 4 (global menu solution and parameter regions). Compute every worker response using Theorem 1. Policy j is globally optimal exactly when, for every feasible i,

$$B_j - B_i + L(I_j - I_i) \geq 0. \tag{15}$$

For fixed behavioral primitives, feasibility cap, and ρ, these inequalities are explicit closed halfspaces in $(D, L, \kappa, \gamma, M)$. Their intersection gives the entire optimality region, not merely a necessary first-order condition. For a one-dimensional sweep in $L \geq 0$, define

$$\ell_j = \max\left\{0, \max_{i:I_j>I_i} \frac{B_i - B_j}{I_j - I_i}\right\},$$

$$u_j = \min_{i:I_j<I_i} \frac{B_i - B_j}{I_j - I_i}. \tag{16}$$

The minimum of an empty set is infinity. A policy with $I_j = I_i$ and $B_j < B_i$ is dominated; otherwise its optimal set is $[\ell_j, u_j]$ if nonempty, with ties retained at endpoints. Along any optimal selection, I is weakly increasing in L. No analogous monotonicity follows for s, a, p, or q.

Unlike a two-regime threshold, (16) tests every feasible alternative. The solution is exact for a finite governance menu, though it does not purport to solve an unrestricted continuous contract problem.

5.2 Why increasing learning value can increase report use

Let i be a lower-audit, more-protected policy and j a higher-audit, less-protected policy. Define $A_j = a_j p_j (1 - q_j)$, the probability mass of audited defects not already reported. Then

$$I_j - I_i = (r_j - r_i) + \rho(A_j - A_i). \tag{17}$$

Corollary 4.1 (audit substitution condition). If $r_j < r_i$ and $A_j > A_i$, policy j supplies more total diagnostic information exactly when

$$\rho > \rho^* = \frac{r_i - r_j}{A_j - A_i}. \tag{18}$$

If additionally $B_i > B_j$, their surplus lines cross at positive $L = (B_i - B_j)/(I_j - I_i)$. They form an actual optimal switch only if all remaining inequalities in (15) hold on the corresponding sides. This final condition is essential: a pairwise crossover alone does not establish governance optimality.

The next theorem supplies all those comparisons, with liability and cost constraints, and verifies a switch toward greater report use. The cost of protection means that greater learning value need not be implemented through a more protected channel if a more informative audit package is selected instead.

5.3 A globally certified, robust counterexample

All primitives in this section are synthetic. Set

$$p_0 = .2,\ c_e = .2,\ k = .3,\ u = .9,\ v = .1,\ \eta = 1,\ h = b = 2,$$

$$D = 3,\ \kappa = .05,\ \gamma = .02,\ \bar{s} = .8,\ B = 2.8,\ M = 0.$$

Use audit choices $\{0, .4, .8\}$ and report-use charges $\{0, .4, .8, 1.2\}$. There are twelve candidates. The liability cap excludes $s = 1.2$ for both positive audit levels, leaving ten feasible policies, including full concealment at $(0, 1.2)$. The primitive bound (6) holds with $G = 1$: $p_0 G/k = 2/3 < 1$ and $h = 2 > 4/3$. Global response uniqueness therefore covers every feasible candidate. We first set $\rho = 1$.

(a, s)	p	q	I	B_j
$(0, 0)$	.091865013	.306216709	.028130602	-.344870749
$(0, .4)$	.087735917	.132042556	.011584875	-.327290266
$(0, .8)$	.086398437	.038447304	.003321787	-.324856871
$(0, 1.2)$	.086174540	.000000000	.000000000	-.327430072
$(.4, 0)$	.075993937	.253313123	.041947732	-.337919030
$(.4, .4)$	.073857422	.111155420	.034468760	-.323564697
$(.4, .8)$	.073126665	.032541366	.030678451	-.321177954
$(.8, 0)$	.065782526	.219275087	.055510915	-.351223990
$(.8, .4)$	.064484615	.097049346	.052839330	-.337810806
$(.8, .8)$	.064028699	.028492771	.051587831	-.335184887

Rounded coefficients are shown only for readability. The executable certificate uses rational primitive inputs and outward interval arithmetic, not the rounded table.

Theorem 5 (nonmonotone optimal protection). For these primitives and $\rho = 1$, define t_m as the exact adjacent-line intersections in the following table. The global optimum over all ten feasible policies is:

Learning-value region	Optimal (a, s)
$0 \le L < t_1$	$(.4, .8)$
$t_1 < L < t_2$	$(.4, .4)$
$t_2 < L < t_3$	$(.8, .8)$
$t_3 < L < t_4$	$(.8, .4)$
$L > t_4$	$(.8, 0)$

At t_m, the two adjacent policies tie. Certified enclosures are

$$t_1 \in [.6296959517914062, .6296959517914065],$$

$$t_2 \in [.6787862980654685, .6787862980654688],$$

$$t_3 \in [2.0982182182734057, 2.0982182182734066],$$

$$t_4 \in [5.020684383434265, 5.020684383434267].$$

In particular, protection increases at t_1 but decreases at t_2: report use rises from .4 to .8 while auditing rises from .4 to .8. Worker reporting sensitivity falls from .11116 to .02849, but total diagnostic information rises from .03447 to .05159 per item. This does not contradict optimal information monotonicity.

The qualitative nonmonotonicity is not confined to perfect audit diagnosis. At the ordered learning-value witnesses $0, .66, 1, 3, 6$, the same five policies are strictly optimal for every $\rho \in [.8, 1]$. Since their strict margins persist by continuity, nonmonotonicity holds on an open neighborhood of other continuous primitives, and around any interior audit quality in this interval. The precise switch thresholds need not equal those tabulated for $\rho = 1$.

For the reversal pair, (18) gives $\rho^* \simeq .2716643$. This is a necessary information-ranking condition for that pair, not a sufficient claim about the full menu at every $\rho > \rho^*$. Tightening the liability cap to $B = 2.4$ excludes $s = .8$ with positive auditing and changes the $L = 0$ optimum to $(.4, .4)$. The liability constraint therefore restricts feasible implementation and can change the optimal policy.

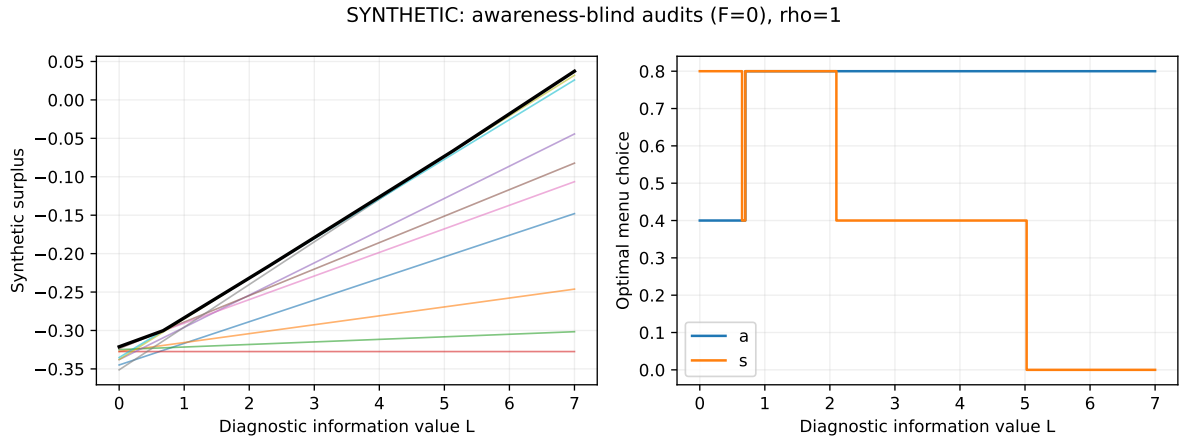


Figure 1: Synthetic global surplus envelope and optimal audit/report-use choices. All policies have F=0.

5.4 Interpretation and welfare limits

The reversal in Theorem 5 requires no accusation of dishonesty from an audit omission. Known charge s affects disclosure, independent occurrence charge b affects prevention, and a higher audit package can supply more diagnostic labels. A protected channel is optimal in some parameter regions and not others. The earlier two-policy comparison is a special pair of inequalities in (15), not the governance rule.

The finite-menu restriction matters. Additional continuous audit choices, alternative reporting rewards, leniency schedules, or outcome-contingent contracts may remove the particular switch or dominate every menu policy. The theorem is a robust existence result within an explicit constrained environment, not a universal comparative static. Its envelope geometry is standard. Its candidate contribution is the economic nonmonotonicity with separately produced information and feasible awareness-blind enforcement.

The learning microfoundation is deliberately bounded to one diagnostic opportunity per independent root cause. Repeated reports about the same cause would saturate value and must not be added linearly. Since defects create opportunities to learn, sufficiently large L may favor policies with more remaining defects; prevention harm is explicitly counted through D, but the assumed future remedy value must be justified in an application. None of the normalized magnitudes is calibrated.

6. Verification and evidence status

The analytical solution is global in worker effort under (6), including $d = 0$, $c = 0$, $c = 1$, and all finite policy boundaries. The policy solver compares all twelve candidates and applies the exact liability rule. It computes the optimality intervals rather than searching only a grid of L values. It brackets every feasible policy's unique effort root with rational endpoints and propagates those brackets using 50-digit interval arithmetic. All nonadjacent inequalities needed for the stated regions are certified; adjacent intersections are enclosed and strictly ordered. Thus the

numerical counterexample is computer-assisted, with executable interval checks, not inferred from a graph.

Independent floating-point checks compare direct three-choice optimization to the analytical response in 126 cases across $F = 0$, hypothetical $F = 1$, policy boundaries, and concealment corners. Finite differences agree with (8) to less than 8×10^{-12} on the checked interior cases. A separate low-remediation example gives exactly zero attention. These checks supplement rather than replace the proof. Package outputs report solver versions and all synthetic inputs.

No real data were used. The measurement appendix distinguishes policy-specific actual defect incidence, reporting sensitivity, and false alarms. It preserves imperfect-audit assumptions and explains the nonidentification of discovery versus concealment without awareness data. The proposed study, participant scripts, dictionary, precision planning, and future-result templates are in a separate protocol, outside the theoretical results. Neither numerical root verification nor a pre-analysis plan supplies empirical validity.

7. Conclusion

Jointly choosing auditing and report use changes the policy conclusion. Without enforceable awareness evidence, the direct fixed-incidence audit-reversal mechanism disappears. Nevertheless, awareness-blind audit design can produce a nonmonotone optimal demand for protection when independent diagnostic information substitutes for worker reports. This is established globally for a constrained menu, with explicit parameter regions and robust strict-winner certificates.

The strongest remaining scientific objection is scope. The menu and cost technology are imposed, continuous and richer contracts are not optimized, and the general investigation-versus-liability tradeoff is well established. The article adds a bounded nonmonotonicity result and an implementability correction, not proof of broad priority or leading-journal suitability. A consequential empirical application would still require independent source artifacts, validated audit quality, credible enforcement of report-use restrictions, and sufficient organizational policy variation. No publication or outside contact has been undertaken.

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Appendix A. Complete proofs

A.1 Global response, including inactive attention

For any discovered defect, the bracketed expression in (2) is strictly concave in c, with derivative $s - v - aF - \eta c$. Maximizing on $[0, 1]$ yields (5). For given e, the attention problem is $\max_{0 \le d \le 1}\{pdg - kd^2/2\}$. It gives $d = [pg/k]_0^1$. Under (6) no positive attention can saturate, so $d = pg_+/k$, including $g \le 0$.

Substituting both choices gives reduced utility

$$\widetilde{U}(e) = -c_e e^2/2 - Hp + p^2 g_+^2/(2k).$$

Its first and second derivatives are

$$\widetilde{U}_e = -c_e e + Hp - p^2 g_+^2/k,$$
$$\widetilde{U}_{ee} = -c_e - Hp + 2p^2 g_+^2/k.$$

Because $p \le p_0$, (6) implies $H - 2pg_+^2/k > 0$, so the second derivative is strictly negative everywhere. At zero effort the first derivative is $p_0(H - p_0 g_+^2/k) > 0$; it tends to minus infinity as effort grows. Exactly one positive maximum therefore solves (7). Attention is then unique. Continuation concealment is unique conditional on a discovery by strict concavity; if $d = 0$ no realized concealment frequency can be inferred. Strict monotonicity of the effort equation and its nonzero derivative give continuous dependence on parameters, including $g = 0$ and concealment corners. Since g_+^2 is differentiable at $g = 0$ with derivative zero, no discontinuity in effort appears there. This proves Theorem 1.

A.2 Feasible audit derivatives and report-use decomposition

With $F = 0$, the continuation optimization (5) has no a dependence. This proves fixed-incidence invariance without any interior assumption: even the truncated attention function is unchanged. Differentiate the effort equation $f = c_e e - Hp + p^2 g_+^2/k = 0$. Its effort derivative is $\Delta > 0$. Since $f_a = -pb$, the implicit function theorem gives $e_a = pb/\Delta$. Therefore $p_a = -pe_a$. With $q = pg_+(1 - c)/k$ and g, c independent of a, $q_a = -e_a q$, including $g_+ = 0$ and $c = 1$. This proves (8) and Theorem 2 at every audit boundary by one-sided differentiation.

Where $0 < c < 1$ and $g > 0$, the envelope theorem gives $g_s = -(1 - c)$ and $c_s = 1/\eta$. Holding e fixed, $f_s = 2p^2 g g_s/k = -2pd(1 - c)$; hence $e_s = 2pd(1 - c)/\Delta$. Differentiating $q = pg(1 - c)/k$ yields (9). For $c = 0$ or $c = 1$, use its zero local derivative; at a switching boundary use adjacent one-sided derivatives. At $g = 0$, q is continuous with the one-sided derivative inherited from the positive-attention branch. In no case does a report omission identify its cause.

For (10), a diagnostic label arrives through a report, or through an audit when no report supplied the label. Conditional independence gives $q + a\rho(1 - q)$. The same argument with $\rho = 1$ gives occurrence detection. Substituting (8) and $p_a = -pe_a$ and differentiating proves (11). This derivation makes the prevention term explicit rather than mixing it with the fixed-incidence derivative.

A.3 Primitive fixed-incidence reversal and every boundary

Fix p and $F > 0$. In the interior concealment region $c = (x - aF)/\eta$, so $c_a = -F/\eta$. Substituting the continuation optimum into g gives $g = u - x + \eta c^2/2$ and $g_a = -Fc$. With positive unsaturated attention,

$$q = pg(1 - c)/k,$$

$$q_a = (p/k)[-Fc(1-c) + gF/\eta] = (pF/k)[(u-x)/\eta + \tfrac{3}{2}c^2 - c].$$

The quadratic is negative strictly between its distinct real roots. Its discriminant is $1 - 6t$, giving (13). One must also require $g > 0$, $0 < c < 1$, unsaturated attention, and a reachable audit intensity. Those restrictions are part of the result; roots alone do not establish a feasible reversal. The minimum of $\frac{3}{2}c^2 - c$ over the real line is $-1/6$, at $c = 1/3$. Hence $t \geq 1/6$ rules out a negative derivative.

If attention is saturated, $q = 1 - c$ and $q_a = F/\eta > 0$ in interior concealment. If $g < 0$, attention and reporting are zero locally. If $c = 0$ on an open interval, $g = u - x$ is independent of a and q is constant. If $c = 1$ on an open interval, $q = 0$ regardless of attention. The truncation maps and continuation value are continuous at $g = 0$, $c = 0$, $c = 1$, and $pg = k$, so the one-sided formulas exhaust their boundaries. When $F = 0$ the continuation problem is independent of audit intensity for every p, including saturation, so all fixed-incidence changes vanish. This proves Theorem 3 globally as a piecewise statement.

A.4 Full policy regions and monotonic information

The finite feasible set is nonempty in the example and, in general, is assumed nonempty. It has a global maximizer because it is finite. For any candidate j, $W_j \geq W_i$ for every i is both necessary and sufficient for global optimality; substitution yields (15). Expanding one inequality gives the explicit halfspace

$$\begin{aligned} D(p_i - p_j) + L(I_j - I_i) + M(T_j - T_i) - \frac{\kappa}{2}(a_j^2 - a_i^2) - \gamma[\phi(s_j) - \phi(s_i)] \\ +u(p_j d_j - p_i d_i) - (E_j - E_i) \geq 0. \end{aligned} \tag{A1}$$

Conditional on worker primitives, all coefficients are fixed. A pair with greater I_j supplies a lower bound on L, one with smaller I_j supplies an upper bound, and equal slopes are decided by intercepts. Intersecting with $L \geq 0$ yields (16), including empty regions and ties. The menu's endpoints need no separate first-order conditions; they are compared directly.

Take $L_2 > L_1$ and optimal policies j_1, j_2. Optimality gives

$$\begin{aligned} B_{j_1} + L_1 I_{j_1} \geq B_{j_2} + L_1 I_{j_2}, \\ B_{j_2} + L_2 I_{j_2} \geq B_{j_1} + L_2 I_{j_1}. \end{aligned}$$

Adding yields $(L_2 - L_1)(I_{j_2} - I_{j_1}) \geq 0$. Thus optimal I is weakly increasing even with ties. The proof does not order any component of the policy pair. Equation (17) follows by expanding (10), and solving its sign inequality under the corollary's assumptions proves (18) and the pairwise crossing claim. This proves Theorem 4 and Corollary 4.1.

A.5 Certificate and robustness proof for Theorem 5

All decimal primitives in Section 5.3 are exact rational numbers. The unique effort for each feasible policy is the root of a strictly increasing $f(e)$, with $f(0) < 0 < f(8)$. The reproduction script performs rational bisection and evaluates the sign of f using outward interval arithmetic. Each nonambiguous sign discards the corresponding half interval without excluding the root. After 120 steps the rational bracket encloses the unique root; propagating its endpoints through p, d, q, E, B, I produces interval coefficients. No floating-point point estimate is used as a proof of a sign.

For adjacent policies j_m, j_{m+1} in the displayed sequence, the exact threshold is

$$t_m = (B_{j_m} - B_{j_{m+1}})/(I_{j_{m+1}} - I_{j_m}).$$

The interval coefficients enclose these roots as reported, with strictly positive denominators and strictly ordered thresholds. For each active policy the script checks every nonadjacent inequality in (16): each competing lower threshold lies strictly below its left active endpoint and each competing upper threshold strictly above its right active endpoint. Adjacent inequalities hold by the definition of t_m. Equal-policy comparisons are identities. This certifies the claimed intervals against all feasible policies for every $L \geq 0$, including both unbounded outer regions. At each endpoint only the adjacent policies tie because all nonadjacent comparisons are strict.

As a second certificate, at $L = 0, .66, 1, 3, 6$, respectively, the minimum surplus margin over every other feasible policy at $\rho = 1$ exceeds \$ .0023867,.0001148,.0013744,.0011285,.0026163\$. The script also certifies strict positivity at $\rho = .8$ for each same witness and every competitor. Differences at a fixed witness are affine in ρ, so endpoint positivity proves positivity on the entire interval $[.8, 1]$. Hence the five optimal choices, including the increase of s from .4 at .66 to .8 at 1, persist for each such audit quality. Worker responses and surplus coefficients are continuous in all the remaining continuous primitives by Theorem 1. Strict finite sets of witness inequalities therefore persist on an open neighborhood; choose an interior ρ, such as .9, for a full relative-interior robustness statement.

The cap $B = 2.8$ places a candidate on its liability boundary. For robustness take caps in $(2.8, 3.2)$, which preserve the same feasible set: $s = .8$ remains feasible at positive a, $s = 1.2$ remains infeasible. Thus there is an ordinary open cap interval as well, rather than incorrectly claiming a two-sided neighborhood around a feasibility discontinuity. For $B = 2.4$ the finite set changes, and direct all-policy comparison at $L = 0$ gives the stated different optimum. These arguments prove Theorem 5 as a computer-assisted existence and parameter-region result.

A.6 Accountability can also reward documentation

The corruption baseline is not a universal prediction. Suppose accountability additionally increases the real private benefit of investigation, $u(s) = u_0 + \omega s$, and independently verified effort incentives enter as $H(s) = h + ab + \chi s$. In an interior concealment region, $g_s = \omega - (1 - c)$ and the same differentiation gives

$$e_s = [p\chi - 2pdg_s]/\Delta,$$
$$q_s = -e_s q + pg_s(1 - c)/k - d/\eta.$$

Thus genuine prevention and sensitivity can rise together if $\chi > 2dg_s$ and $pg_s(1 - c)/k > e_s q + d/\eta$. These conditions follow directly from the derivatives and are not field estimates. At concealment corners omit the d/η term locally. This alternative changes worker primitives and therefore requires recomputing every policy region; it is not silently built into the certified baseline.

Appendix B. Measurement and limits of identification

The operational unit is a completed item and policy can be assigned by team. Define $p_t = \Pr(Z(t) = 1)$, $q_t = \Pr(R(t) = 1 \mid Z(t) = 1)$, and $f_t = \Pr(R(t) = 1 \mid Z(t) = 0)$. They satisfy

$$\Pr(R(t) = 1) = p_t q_t + (1 - p_t) f_t.$$

A report count alone does not separate these quantities. Moreover, q_t concerns each policy's own defect population. A contrast in q_t is not a behavioral effect among the latent items defective under both policies. Severity-standardized comparisons are secondary and do not identify that principal stratum.

Let audit selection have known positive inclusion probability π_i. Probability weighting identifies the joint table of auditor classification Y and recording R under complete follow-up or an explicitly justified missingness model. Let validated audit sensitivity and specificity be α, β, and assume $Y \perp R \mid Z, t, X$. For each report status j,

$$\begin{pmatrix} P(Y=0, R=j) \\ P(Y=1, R=j) \end{pmatrix} = \begin{pmatrix} \beta & 1-\alpha \\ 1-\beta & \alpha \end{pmatrix} \begin{pmatrix} P(Z=0, R=j) \\ P(Z=1, R=j) \end{pmatrix}. \tag{B1}$$

The matrix determinant is $\alpha+\beta-1$. If this is nonzero and the error rates are known, inversion identifies the latent joint table; summing its true-defect row gives p, dividing its $(Z=1, R=1)$ cell by p gives q, and dividing its $(Z=0, R=1)$ cell by $1-p$ gives f. Sensitivity is undefined at $p=0$, false alarms at $p=1$. The usual informative-audit restriction is $\alpha+\beta>1$. If error rates vary by report status, treatment, or covariate stratum, each required matrix must be validated separately.

This is standard misclassification algebra, not a new estimator. Negative inverse cells in samples are a warning, not an invitation to truncate silently. Use a constrained likelihood and propagate validation uncertainty. Bounds over feasible latent tables and validated error ranges replace point estimates when accuracy is uncertain. Two reviewers of the same incomplete source do not automatically create a gold standard. Missing audits and nonrandom selection require explicit adjustment assumptions or bounds. Organizational auditing capability ρ is not estimated by merely counting research-audit disagreements: diagnostic labels need their own validation.

Without observed awareness, $q = d(1-c)$ does not identify its components. For any $q \in (0,1)$, $(d,c) = (q,0)$ and $(1, 1-q)$ both reproduce it. Occurrence audits, even perfect ones, cannot discriminate these behaviors without additional information or exclusion restrictions. A timestamped draft can establish awareness in a subset but does not exhaust all awareness; conditioning on drafts selects a post-treatment subset. This nonidentification remains even though the theoretical worker model has separate choices. The empirical protocol must therefore estimate policy-specific reporting sensitivity, not claim a causal decomposition into discovery and dishonesty.